OOP Activity 4

Title: Write a program in C++ to perform arithmetic operations on complex numbers. Design the class for complex number representation and the operations to be performed. The objective of this assignment is to learn the concepts of classes and objects.		
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CODE

```
#include <iostream>
using namespace std;

class complex{
    float real1, real2, imaginary1, imaginary2;

    public:
    void getData() {
        cout << "First Number: " << endl;
        cout << "Enter real part: ";
        cin >> real1;
        cout << "Enter imaginary part: ";
        cin >> imaginary1;
        cout << endl << "Second number: " << endl;
        cout << "Enter real part: ";
        cin >> real2;
```

```
        cout << "Enter imaginary part: ";
        cin >> imaginary2;
    }


    void addData() {
        int realSum = real1 + real2;
        int imgSum = imaginary1 + imaginary2;
        cout << "Sum is: " << realSum;

        if(imgSum>0)
        cout << " + " << imgSum << "i";
        else if(imgSum<0)
        cout << " - " << -imgSum << "i";
    }


    void subtractData() {
        int realSub = real1 - real2;
        int imgSub = imaginary1 - imaginary2;
        cout << "Difference is: " << realSub;

        if(imgSub>0)
        cout << " + " << imgSub << "i";
        else if(imgSub<0)
        cout << " - " << -imgSub << "i";
    }
```

```
    void multiplyData() {
        int realTerm = (real1*real2) -
(imaginary1*imaginary2);
        int imaginaryTerm = (real1*imaginary2) +
(real2*imaginary1);
        cout << "Product is: " << realTerm;

        if(imaginaryTerm>0)
        cout << " + " << imaginaryTerm << "i";
        else if(imaginaryTerm<0)
        cout << " - " << -imaginaryTerm << "i";
    }


    void divideData() {
        float realTerm1 = ((real1*real2) +
(imaginary1*imaginary2)) /
((real2*real2)+(imaginary2*imaginary2));
        float imaginaryTerm1 = ((real2*imaginary1) -
(real1*imaginary2)) /
((real2*real2)+(imaginary2*imaginary2));
        cout << "Division is: " << realTerm1;

        if(imaginaryTerm1>0)
        cout << " + " << imaginaryTerm1 << "i";
```

```
        else if(imaginaryTerm1<0)
        cout << " - " << -imaginaryTerm1 << "i";
    }
};

int main() {

    complex c1;
    c1.getData();
    cout << endl;
    c1.addData();
    cout << endl;
    c1.subtractData();
    cout << endl;
    c1.multiplyData();
    cout << endl;
    c1.divideData();


    return 0;
}
```

OUTPUT
First Number: Enter real part: 8 Enter imaginary part: 6 Second number: Enter real part: -4 Enter imaginary part: 6 Sum is: 4 + 12i Difference is: 12 Product is: -68 + 24i Division is: 0.0769231 - 1.38462i